Final Exam Review

L1 - 29, 50/50 old/new
L1-24 L25-29

1st Order : Euler's Method

Separable, Exact, Integrating factors, Linear, homogeneous, Bernoulli, substitutions

Ex: $\frac{dy}{dx} + \underbrace{\frac{y}{x}}_{p(x)=\frac{1}{x}} = \underbrace{x^2}_{q'(x)}\underbrace{y^3}$

Identify : **Bernoulli**

$y' + py = q = q'y^n$

$\uparrow$ of x

$\Rightarrow v' + (1-n)pv = (1-n)q'$

$n = 3$

$v = y^{1-n} = y^{-2} \rightsquigarrow$

$\Rightarrow v' + (-2)\frac{1}{x}v = (-2)x^2$ (Linear)

$\Rightarrow v' - \frac{2}{x}v = -2x^2$

$y' + py = q$ linear

$\mu = e^{\int p\,dx}$

$\Rightarrow (\mu y)' = \mu q$

$\Rightarrow \mu = e^{\int -\frac{2}{x}dx}$

$= e^{-2\ln|x|} = x^{-2}$

$\Rightarrow \int (x^{-2}v)' = \int x^{-2}(-2x^2)$

$\Rightarrow x^{-2}v = \int -2\,dx + C = -2x + C$

$\Rightarrow v = -2x^3 + Cx^2$, $v = y^{-2}$

$\Rightarrow y^{-2} = -2x^3 + Cx^2$ (implicit)

$\Rightarrow \boxed{y^2 = \frac{1}{-2x^3 + Cx^2}}$

Ex: $(y-4x-1)^2 dx - dy = 0$

Identify:

separable ✗

linear ✗

exact $M\,dx + N\,dy = 0$ ✗

$M_y = N_x$?

true then exact!

$M = (y-4x-1)^2 \Rightarrow M_y = 2(y-4x-1)$

$N = -1 \qquad \Rightarrow N_x = 0$ $\neq$

$\frac{dy}{dx} = (y-4x-1)^2$

$= G(ax+by+c)$

separable after sub.

Substitution?

$z = ax+by+c$

$\boxed{z = y-4x-1}$

$\frac{dz}{dx} = \frac{dy}{dx} - 4$

$\Rightarrow dz = dy - 4dx$

$\Rightarrow dz + 4dx = dy$

Now,

$(y-4x-1)^2 dx - dy = 0 \Rightarrow z^2 dx - (dz + 4dx) = 0$

$\Rightarrow (z^2-4)\,dx - dz = 0$

$\Rightarrow dz = (z^2-4)\,dx$

$\Rightarrow \int \frac{1}{z^2-4}\,dz = \int dx$

$\Rightarrow \int \frac{1}{(z-2)(z+2)}\,dz = x + C$

$\frac{1}{(z-2)(z+2)} = \frac{A}{z-2} + \frac{B}{z+2} \Rightarrow 1 = A(z+2) + B(z-2)$

$z = -2 \Rightarrow B = -\frac{1}{4}$

$z = 2 \Rightarrow A = \frac{1}{4}$

$$\Rightarrow \int \left(\frac{1/4}{z-2} - \frac{1/4}{z+2} \right) dz = x + C$$

$$\Rightarrow \frac{1}{4} \ln|z-2| - \frac{1}{4} \ln|z+2| = x + C$$

$$\Rightarrow \frac{1}{4} \ln\left| \frac{z-2}{z+2} \right| = x + C$$

$$\Rightarrow \ln\left(\left(\frac{z-2}{z+2}\right)^{1/4} \right) = x + C$$

$$\Rightarrow \left(\frac{z-2}{z+2}\right)^{1/4} = Ce^{x}$$

$$\Rightarrow \frac{z-2}{z+2} = Ce^{4x}$$

$$\Rightarrow \frac{(y-4x-1)-2}{y-4x-1+2} = Ce^{4x}$$

$$\Rightarrow y - 4x - 3 = Ce^{4x}(y - 4x + 1)$$

$$\Rightarrow y(1 - Ce^{4x}) = 4x + 3 - 4Ce^{4x} \cdot x + Ce^{4x}$$

$$\Rightarrow \boxed{y = \frac{4x + 3 - 4Ce^{4x}x + Ce^{4x}}{1 - Ce^{4x}}}$$

Ex: $\theta\, dy - y\, d\theta = \sqrt{\theta y}\, d\theta$

Identify:

separable? ×

linear? ×

exact? ×

$(\underbrace{\sqrt{\theta y} + y}) d\theta - \theta\, dy$

$x = \theta$

$M_y = \frac{1}{2}\frac{\sqrt{\theta}}{\sqrt{y}} + 1$

$N_\theta = -1$ ×

becomes separable

This is homogeneous: divide by θ

$\Rightarrow dy - \frac{y}{\theta}\, d\theta = \frac{\sqrt{\theta y}}{\theta}\, d\theta$

$\Rightarrow dy - \frac{y}{\theta}\, d\theta = \sqrt{\frac{y}{\theta}}\, d\theta$

use $u = \frac{y}{\theta}$

$\Rightarrow y = u\theta$

$\Rightarrow dy = u\, d\theta + \theta\, du$

$\Rightarrow (\cancel{u\, d\theta} + \theta\, du) \cancel{- u\, d\theta} = \sqrt{u}\, d\theta$

$\Rightarrow \theta\, du = \sqrt{u}\, d\theta$

$\Rightarrow \int \frac{1}{\sqrt{u}}\, du = \int \frac{1}{\theta}\, d\theta \Rightarrow 2\sqrt{u} = \ln|\theta| + C$

$\Rightarrow 2\sqrt{\frac{y}{\theta}} = \ln|\theta| + C$ (implicit)

$\Rightarrow \frac{y}{\theta} = \left(\frac{\ln|\theta| + C}{2}\right)^2$

$\Rightarrow \boxed{y = \theta\left(\frac{\ln|\theta| + C}{2}\right)^2}$

Ex: Solve $\frac{dy}{dx} - 4y = 32x^2$ linear

$P(x) = -4$

$Q(x) = 32x^2$

Here: $\mu = e^{\int -4\,dx} = e^{-4x}$

$\Rightarrow (e^{-4x}y)' = e^{-4x} \cdot 32x^2$

$\Rightarrow e^{-4x}y = \int 32x^2 e^{-4x}\,dx + C$

$\Rightarrow e^{-4x}y = (-8x^2e^{-4x} - 4xe^{-4x} - e^{-4x}) + C$

$\Rightarrow \boxed{y = (-8x^2 - 4x - 1) + Ce^{4x}}$

IBP

x^2	$+\ 32e^{-4x}$
$2x$	$-8e^{-4x}$
2	$+\ 2e^{-4x}$
0	$-\frac{1}{2}e^{-4x}$

Ex: $(2y^2x - y)\,dx + x\,dy = 0$ exact? X

$M_y = 4yx - 1$

$\neq$

$N_x = 1$

$2y^2x - y = y(2yx - 1)$

If we want to see if there is an integrating factor of just x or y:

$\frac{M_y - N_x}{N}$ of x

or $\frac{N_x - M_y}{M}$ of y

$\frac{N_x - M_y}{M} = \frac{1-(4yx-1)}{2y^2x - y}$

$$\Rightarrow \frac{N_x - M_y}{M} = \frac{2-4yx}{2y^2x-y} = \frac{-2(\cancel{2yx-1})}{y(\cancel{2yx-1})} = -\frac{2}{y} \checkmark$$

$$\text{So } \mu(y) = e^{\int \frac{N_x-M_y}{M} dy} = e^{\int -\frac{2}{y} dy} = e^{-2\ln|y|} = y^{-2}$$

$$y^{-2}\left[(2y^2x - y)dx + x\,dy = 0\right]$$

$$\Rightarrow (2x - y^{-1})dx + xy^{-2}dy = 0$$

new $\begin{pmatrix} M_y = y^{-2} \\ N_x = y^{-2} \end{pmatrix}$ ✓

there exists a function $f(x,y)$ such that

$$\begin{cases} f_x = M \\ f_y = N \end{cases} \Rightarrow f = \int M dx = \int (2x - y^{-1})dx = x^2 - xy^{-1} + g(y)$$

or ~~$f = \int N dy$~~

$$\Rightarrow f = x^2 - xy^{-1} + g(y)$$

next

$$xy^{-2} = N = f_y = \frac{\partial}{\partial y}(x^2 - xy^{-1} + g(y))$$

$$= 0 - x(-y^{-2}) + g'(y)$$

$$\Rightarrow xy^{-2} = xy^{-2} + g'(y) \Rightarrow g'(y) = 0$$

$$\Rightarrow g(y) = C$$

$$\text{So } f(x,y) = \boxed{x^2 - xy^{-1} = C}$$

Ex: Consider $4y'' - 12y' + 9y = \underbrace{e^{5t}}_{r=5} + \underbrace{e^{3t}}_{r=3}$

use Undetermined Coefficients. $\neq 3/2$

$$4(r-\tfrac{3}{2})^2 = 0$$

$=$

① Solve $4y'' - 12y' + 9y = 0 \Rightarrow 4r^2 - 12r + 9 = 0$

$$\Rightarrow r = \frac{12 \pm \sqrt{144 - 4(9)(4)}}{8}$$

$$= \frac{12 \pm \sqrt{\cancel{144 - 144}}}{8}$$

$$= \frac{12}{8} = \frac{3}{2} \text{ repeated}$$

$$y_h = c_1 e^{3/2 t} + c_2 t e^{3/2 t}$$

② $f = e^{5t} + e^{3t}$, $r = 5, 3 \neq \frac{3}{2}$

use $y_p = Ae^{5t} + Be^{3t}$

$$y_p' = 5Ae^{5t} + 3Be^{3t}$$

$$y_p'' = 25Ae^{5t} + 9Be^{3t}$$

In $4y'' - 12y' + 9y = e^{5t} + e^{3t}$

$$\Rightarrow 4(25Ae^{5t} + 9Be^{3t}) - 12(5Ae^{5t} + 3Be^{3t}) + 9(Ae^{5t} + Be^{3t})$$

$$= e^{5t} + e^{3t}$$

$$\Rightarrow (100A - 60A + 9A)e^{5t} + (36B - 36B + 9B)e^{3t}$$

$$= e^{5t} + e^{3t}$$

$$\Rightarrow 49Ae^{5t} + 9Be^{3t} = e^{5t} + e^{3t}$$

$$\Rightarrow \begin{cases} 49A = 1 \\ 9B = 1 \end{cases} \Rightarrow A = \frac{1}{49}, \quad B = \frac{1}{9}$$

So $$y_g = y_h + y_p = \left(c_1 e^{3/2 t} + c_2 t e^{3/2 t}\right) + \left(\frac{1}{49} e^{5t} + \frac{1}{9} e^{3t}\right)$$

Final Exam Review Continued

2nd order Equation Techniques

Ex: $1 \cdot y'' + 16y = \tan(4t) \leftarrow f$ $\quad$ $y_h = c_1 y_1 + c_2 y_2$

$\uparrow a$

Variation of Parameters $\quad$ $y_p = v_1 y_1 + v_2 y_2$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, \quad W_1 = \begin{vmatrix} 0 & y_2 \\ f/a & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f/a \end{vmatrix}$$

then $\quad v_1 = \int \frac{W_1}{W} dt, \quad v_2 = \int \frac{W_2}{W} dt$

① $y'' + 16y = 0 \Rightarrow r^2 + 16 = 0 \Rightarrow r = \pm 4i$ $\quad$ $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

$$\Rightarrow y_h = c_1 \underbrace{\cos(4t)}_{y_1} + c_2 \underbrace{\sin(4t)}_{y_2}$$

② $$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos(4t) & \sin(4t) \\ -4\sin(4t) & 4\cos(4t) \end{vmatrix}$$

$$= 4\cos^2(4t) - (-4\sin^2(4t)) = 4$$

$4(\sin^2(4t) + \cos^2(4t))$

here $f/a = \frac{\tan(4t)}{1} = \tan(4t)$

$$W_1 = \begin{vmatrix} 0 & \sin(4t) \\ \tan(4t) & 4\cos(4t) \end{vmatrix} = 0 - \sin(4t)\tan(4t)$$
$$= -\sin(4t)\tan(4t)$$

$$W_2 = \begin{vmatrix} \cos(4t) & 0 \\ -4\sin(4t) & \tan(4t) \end{vmatrix} = \cos(4t)\tan(4t) - 0$$
$$= \sin(4t)$$

③ Now $v_1 = \int \frac{W_1}{W}\,dt = \int \frac{-\sin(4t)\tan(4t)}{4}\,dt$

$$= \int \frac{-\sin(4t)\,\frac{\sin(4t)}{\cos(4t)}}{4}\,dt = \int -\frac{1}{4}\cdot\frac{\sin^2(4t)}{\cos(4t)}\,dt$$

$$= \int -\frac{1}{4}\cdot\frac{(1-\cos^2(4t))}{\cos(4t)}\,dt = \int -\frac{1}{4}\cdot\left(\frac{1}{\cos(4t)} - \frac{\cos^2(4t)}{\cos(4t)}\right)dt$$

$$= \int -\frac{1}{4}\left(\sec(4t) - \cos(4t)\right)dt$$

$$= -\frac{1}{4}\left(\frac{1}{4}\ln|\sec(4t)+\tan(4t)| - \frac{1}{4}\sin(4t)\right)$$

$$= \frac{1}{16}\left(\sin(4t) - \ln|\sec(4t)+\tan(4t)|\right)$$

$$v_2 = \int \frac{W_2}{W}\,dt = \int \frac{\sin(4t)}{4}\,dt = -\frac{1}{16}\cos(4t)$$

So $y_g = y_h + y_p = (c_1 y_1 + c_2 y_2) + (v_1 y_1 + v_2 y_2)$

$$= \left(c_1\cos(4t) + c_2\sin(4t)\right) + \left(\frac{1}{16}\left(\sin(4t) - \ln|\sec(4t)+\tan(4t)|\right)\right.$$

$$\left.\cdot\,(\cos(4t)) + \left(-\frac{1}{16}\cancel{\cos(4t))(\sin(4t)}\right)\right)$$

$$= (c_1\cos(4t) + c_2\sin(4t)) + \left(-\frac{1}{16}\ln|\sec(4t)+\tan(4t)|\cos(4t)\right)$$

Ex: Consider $ty'' + (1-2t)y' + (t-1)y = 0, \quad t > 0$

Given: $y_1 = e^t$ is a solution, find y_2.

std form

Reduction of Order: $y_2 = y_1 \int \frac{e^{\int -P\,dt}}{y_1^2}\,dt$

$\Rightarrow y'' + \underbrace{\frac{(1-2t)}{t}}_{P} y' + \underbrace{\frac{(t-1)}{t}}_{q} y = 0$

$$y_2 = e^t \int \frac{e^{\int -\frac{(1-2t)}{t}\,dt}}{(e^t)^2}\,dt$$

$$= e^t \int \frac{e^{\int (-\frac{1}{t} + 2)\,dt}}{e^{2t}}\,dt = e^t \int \frac{e^{-\ln(t) + 2t}}{e^{2t}}\,dt$$

$$= e^t \int \frac{t^{-1} e^{2t}}{e^{2t}}\,dt = e^t \int t^{-1}\,dt = e^t \ln(t)$$

Laplace Transforms

Ex: Solve $\quad y + \int_0^t \underset{f(t)=t}{(t-v)}\,\underset{g(t)=y}{y(v)}\,dv = e^{-3t}$

Convolution: $(f*g)(t) = \int_0^t f(t-v)\,g(v)\,dv$

$$\mathcal{L}\{f*g\} = F(s)\,G(s)$$

$$\Rightarrow \quad y + (t * y) = e^{-3t}$$

apply the transform: $Y(s) + \frac{1}{s^2} \cdot Y(s) = \frac{1}{s+3}$

$$\Rightarrow \quad Y(s)\left(1 + \frac{1}{s^2}\right) = \frac{1}{s+3}$$

$$\Rightarrow \quad Y(s)\left(\frac{s^2+1}{s^2}\right) = \frac{1}{s+3}$$

$$\Rightarrow \quad Y(s) = \frac{1}{s+3} \cdot \frac{s^2}{s^2+1} = \frac{s^2}{(s+3)(s^2+1)}$$

PFD:

$$\frac{s^2}{(s+3)(s^2+1)} = \frac{A}{s+3} + \frac{Bs+C}{s^2+1} \Rightarrow s^2 = A(s^2+1) + (Bs+C)(s+3)$$

$s=-3$: $9 = A(9+1) \Rightarrow A = \frac{9}{10}$

$s=0$: $0 = \frac{9}{10}(0+1) + (C)(3) \Rightarrow 3C = -\frac{9}{10}$

$$\Rightarrow C = -\frac{3}{10}$$

$s=1$: $1 = \frac{9}{10}(1+1) + \left(B - \frac{3}{10}\right)(4)$

$$\Rightarrow 4B - \frac{12}{10} + \frac{18}{10} = 1 \Rightarrow 4B = 1 - \frac{3}{5} = \frac{2}{5}$$

$$\Rightarrow B = \frac{1}{10}$$

So $Y(s) = \dfrac{9/10}{s+3} + \dfrac{\frac{1}{10}s - \frac{3}{10}}{s^2+1}$

$$\Rightarrow y(t) = \frac{9}{10}e^{-3t} + \frac{1}{10}\cos(t) - \frac{3}{10}\sin(t)$$

Ex: $\mathcal{L}^{-1}\left\{\dfrac{e^{-2s}(4s+2)}{(s-1)(s+2)}\right\} = \mathcal{L}^{-1}\left\{e^{-2s}\left(\dfrac{4s+2}{(s-1)(s+2)}\right)\right\}$

$$\frac{4s+2}{(s-1)(s+2)} = \frac{A}{s-1} + \frac{B}{s+2} \Rightarrow 4s+2 = A(s+2) + B(s-1)$$

$s=1$: $A=2$

$s=-2$: $B=2$

$$\mathcal{L}^{-1}\left\{e^{-2s}\left(\frac{2}{s-1}+\frac{2}{s+2}\right)\right\} = u(t-2)\left[2e^{t}+2e^{-2t}\right]\Big|_{t\to t-2}$$

$$\boxed{\begin{gathered}\mathcal{L}^{-1}\{e^{-as}F(s)\} = u(t-a)f(t-a)\\ \mathcal{L}\{u(t-a)f(t)\} = e^{-as}\mathcal{L}\{f(t+a)\}\end{gathered}}$$

$$= u(t-2)\left(2e^{t-2}+2e^{-2(t-2)}\right)$$

<u>Euler's Method</u> $\quad \dfrac{dy}{dx} = f(x,y)\,, \quad y(x_0)=y_0$

Given a step size h then

$$\begin{cases} x_i = x_{i-1} + h \\ y_i = y_{i-1} + h f(x_{i-1}, y_{i-1}) \end{cases} \qquad i \geq 1$$

$\underline{Ex}$: Consider $\frac{dy}{dx} = \underbrace{2x-y}_{f(x,y)}$, $y(\underset{x_0}{0})=\overset{y_0}{1}$, $h=0.1$ find y_2.

$$\begin{cases} x_1 = x_0 + h = 0 + 0.1 = 0.1 \\ y_1 = y_0 + hf(x_0,y_0) = 1 + 0.1f(0,1) \\ \quad = 1 + 0.1(2\cdot 0 - 1) = 1 - 0.1 = 0.9 \end{cases}$$

$$\begin{cases} x_2 = 0.2 \\ y_2 = y_1 + h f(x_1,y_1) = 0.9 + 0.1(2(0.1) - 0.9) \\ \quad = 0.9 + 0.1(-0.7) \\ \quad = 0.9 - 0.07 = \boxed{0.83} \approx y(0.2) \end{cases}$$

<u>Phase - Plane</u>

Ex:
$$\begin{cases} x' = y^2 - 3y + 2 \\ y' = (x-1)(y-2) \end{cases}$$

find the critical points:
$$\begin{cases} y^2 - 3y + 2 = 0 \Rightarrow (y-2)(y-1) = 0 \\ (x-1)(y-2) = 0 \end{cases}$$

$y = 1, 2$

$\underline{y=1}$

$(x-1)(1-2) = 0$

$-(x-1) = 0$

$x = 1$

$(1,1)$

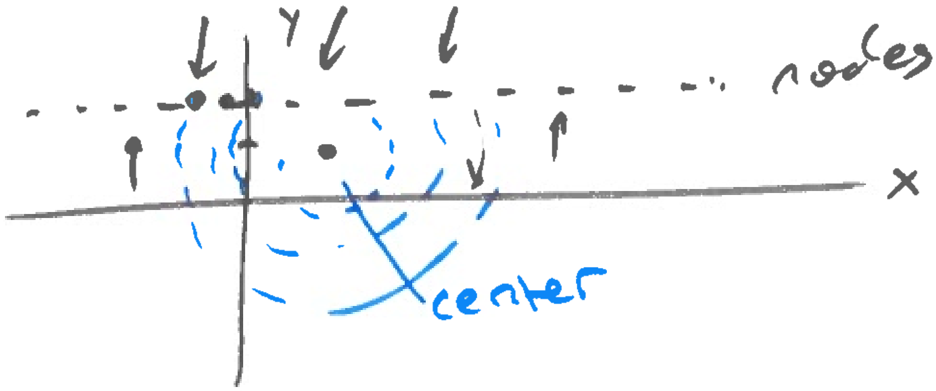
$\underline{y=2}$

$(x-1)(2-2) = 0$

$0 = 0$

true for all values of x

line $\underline{y=2}$



Ex: Consider
$$\begin{cases} x' = \dfrac{3}{y} \\ y' = \dfrac{2}{x} \end{cases}$$

Solve the phase-plane equation

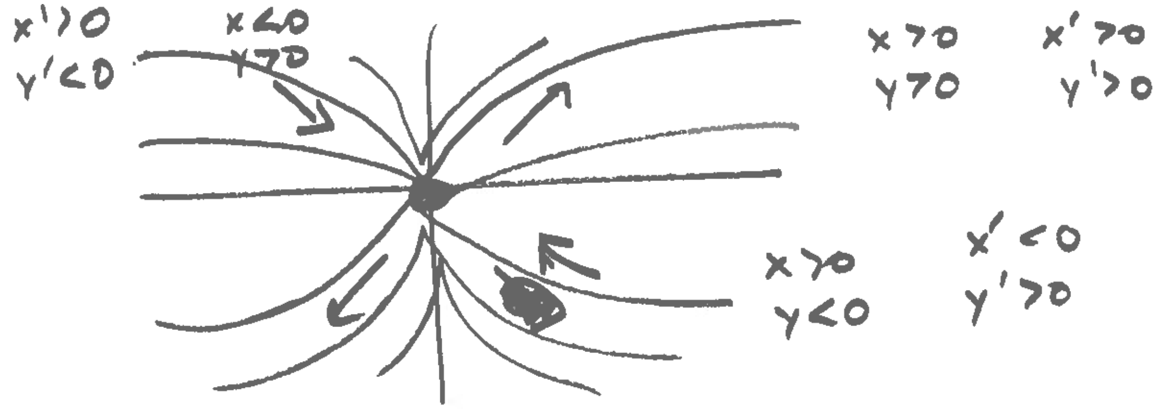
$$\frac{dy}{dx} = \frac{y'}{x'} = \frac{\frac{2}{x}}{\frac{3}{y}} = \frac{2y}{3x} \quad \text{separable}$$

$$\Rightarrow \int \frac{1}{2y}\,dy = \int \frac{1}{3x}\,dx$$

$$\Rightarrow \frac{1}{2}\ln|y| = \frac{1}{3}\ln|x| + C$$

$$\Rightarrow \ln|y^{1/2}| = \ln|x^{1/3}| + C$$

$$\Rightarrow y^{1/2} = Cx^{1/3} \quad \text{or} \quad y = Cx^{2/3}$$



saddle.